

Chebyshev-Halley exclusion by incompatible coefficients

$$6(3 - 2\alpha) = 15 - 6\alpha \rightarrow \alpha = 1/2$$

$$12\alpha = 4 + 2\alpha \rightarrow \alpha = 2/5$$

one parameter cannot be both $2/5$ and $1/2$

Chebyshev
 $\alpha=0$

forced by
 $s^6: 1/2$

super-Halley
 $\alpha=1$

forced by
 $s^3 X: 2/5$

Lean conclusion: no α satisfies both matching equations.

